

ADITYA PATIL

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SUMMARY

Cybersecurity professional and M.S. candidate with four years plus hands-on experience in penetration testing, vulnerability management, product security, and external attack surface assessment. Experienced in identifying and validating vulnerabilities across web, API, IoT, mobile, and LLM-integrated systems and large-scale organizations through manual testing, source code review, reverse engineering, and OSINT.

EXPERIENCE

Security Researcher - Rochester Institute of Technology, SAFE Lab Jan. 2025 - Present

Project 1:

- Performed penetration testing on an IoT device for an enterprise client.
- Exploited security misconfigurations to break out of kiosk mode on an IoT device.
- Wrote scripts for exploitation with BadUSB.
- Reverse engineered underlying software. Identified security issues and cryptographic failures.
- Uncovered issues in authentication mechanism with source code review and gained access to Azure Cloud tenant.
- Communicated 5 major security gaps and findings with technical and executive audience through effective reporting.

Project 2:

- Performed OSINT and external attack surface mapping on four large scale entities for a vendor.
- Carried out reconnaissance activity to discover web application assets.
- Gathered personnel intel as part of OSINT.
- Discovered major security misconfigurations and information disclosure issues.
- Communicated 5 major security gaps and 39 findings across four entities with executive audience through effective report writing and video presentation.

Project 3:

- Performed OSINT and external attack surface mapping for an enterprise client.
- Discovered web application assets and mobile applications.
- Reverse engineered iOS applications.
- Discovered TLS Pinning, information disclosure and cryptographic material exposure.
- Communicated 4 major security gaps and 7 findings across four entities with executive audience through effective report writing and video presentation.

Vulnerability Management Co-op - Rochester Regional Health Aug. 2025 - July 2026

- Planned and executed pentests on 6+ internal and external web/API applications.
- Performed third party risk evaluations and security architect reviews and identified major security gaps.
- Regularly monitored external attack surface of 5 main subsidiary organizational units.
- Ran DAST scans on multiple applications.
- Worked with application and infrastructure team remediate security issues.

Teaching Assistant - Rochester Institute of Technology Jan. 2025 - May 2025

- Facilitated hands-on labs and tutoring using VMware Workstation Pro, pfSense, Windows Server 2022, Windows 10, and Rocky Linux, covering virtual infrastructure, LAN segmentation, NAT, static IPv4 addressing, DNS, DHCP, and troubleshooting.
- Supported instruction and assessment in Active Directory Domain Services, Group Policy Objects, domain joining, organizational units, user and group administration, OpenLDAP, Kerberos authentication, PowerShell, and Windows Admin Center.
- Guided students through Linux user and permission management, file systems, GPT/MBR partitioning, RAID, NAS/SAN storage, cron automation, Git, and security fundamentals including access control, encryption, TLS, and authentication.

Systems Engineer (Cybersecurity) - Tata Consultancy Services June 2023 - June 2024

- Identified and reported security vulnerabilities across 25+ web and API assets for a Big Four client, including cross site scripting (XSS), cross site request forgery (CSRF), SQL injection (SQLi), privilege escalation and sensitive data disclosures.
- Conducted manual penetration tests using Burp Suite and validated risks through exploitation.
- Performed pentests on LLM-integrated applications and reported multiple OWASP Top 10 vulnerabilities.
- Automated reconnaissance workflows using custom Python scripts, saving ~ 5 hours per pentest.

Security Consultant Intern - SecureLayer7 March 2023 - May 2023

- Performed web and API penetration testing for enterprise clients; discovered critical issues including XSS, CSRF, SQLi, and privilege escalation and delivered clear, actionable reports with OWASP & PTES standards.

Project Intern - Wipro March 2022 - June 2022

- Developed a full-stack MERN application, implementing a RESTful API with Node.js/Express and a responsive React frontend to perform full CRUD operations.

AWARDS & ACHIEVEMENTS

- 2nd Place - Collegiate Penetration Testing Competition (CPTC) Great Lake Regional 2025.
- Finalist - Collegiate Penetration Testing Competition (CPTC) Global Finals 2026.
- Top 100 out of 12000 candidates - TCS HackQuest Season 7 CTF Competition.
- Hall of Fame - Drexel University & State of California - reported critical vulnerabilities exposing PII & PHI

EDUCATION

Rochester Institute of Technology | M.S. Cybersecurity
Fall 2024 - Expected Dec. 2026 • GPA: 3.89/4.00

Savitribai Phule Pune University | B. E. Computer engineering
June 2018 - April 2022 • GPA: 8.95/10.00

PROJECTS

Linux Package Integrity & Authentication using a PQC Hybrid RSA-2048 and ML-DSA-65

- Developed a hybrid-signature integrity-verification system for Linux packages using RSA-2048 and post-quantum ML-DSA-65 to resist quantum attacks and ensure authenticity.

Malicious URL Detection Using Machine Learning

- Developed a logistic regression-based model to detect malicious URLs based on lexical analysis.

SKILLS & TOOLS

Pentesting: Burp Suite, Caido, IBM AppScan, Metasploit, Nmap, Wireshark, Postman

Vulnerability Management: Rapid7 InsightVM, InsightAppSec,

Programming: Python, C, C++, JavaScript

Security: Web, API & IoT Security, PTES, OWASP, SAST/DAST, Thick Client & LLM Security Testing

Coursework: Cryptography, Cyber Analytics & Machine Learning, Adv. Malware & Computer Forensics